YuEEG / REV E / DUAL USB-C / AMOLED + SD BACKUP / FIXED AFE

01 / Electrode inputs CH1-CH4

File: inputs_1_4.kicad_sch

01 / Electrode inputs CH5-CH8

File: inputs_5_8.kicad_sch

02 / ADS1299 digital, bias, reference and decoupling

File: afe_core.kicad_sch

03 / ESP32-S3 wireless acquisition controller

File: mcu_radio.kicad_sch

04 / Battery input and independent analog-digital supplies

File: power.kicad_sch

05 / Bipolar analog supply

File: power_bipolar.kicad_sch

06 / USB-C standalone 2S charger and battery NTC

File: battery_charger.kicad_sch

07 / Raw 2S cell protection and fail-safe acquisition switch

File: battery_protection.kicad_sch

08 / Cell balancing with raw-domain UV qualification

File: battery_balance.kicad_sch

09 / Isolated USB data and conditional isolated power

File: isolated_usb.kicad_sch

10 / USB SYS power path with delayed battery interlock

File: usb_system_power.kicad_sch

12 / AMOLED 1.91 inch RM67162 - DWO DO0200PFS12

File: amoled_ui.kicad_sch

13 / SDMMC backup, status and low-profile controls

File: storage_status.kicad_sch

14 / Separate USB-C DATA and CHARGE primary-side ports

File: dual_usb_ports.kicad_sch

Engineering review. Screen, storage, USB, battery and patient isolation require physical validation.
Separate DATA and CHARGE ports; no external analog expansion. Follow the Rev E assembly notes.

DUAL USB / AMOLED / SD BACKUP - ENGINEERING VALIDATION REQUIRED

YuEEG

Sheet: /

File: YuEEG_8CH_RevE.kicad_sch

Title: YuEEG 8CH - wireless EEG acquisition

Size: A3	Date: 2026-10-27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 1/15

01 / ELECTRODE INPUTS CH1–CH4

Referential EEG: one SIGNAL terminal per channel. Shared REF –> all N input networks. BIAS is a separate driven electrode.

CH1 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J101
CH1 EEG
E1P
1

R111
4.99k 0.1%
E1P
CLAMP1P
CLAMP1P
AIN1P

R112
4.99k 0.1%

R113
4.99k 0.1%
E1N
CLAMP1N
CLAMP1N
AIN1N

R114
4.99k 0.1%

R115
0R REF SELECT
REF_RAW
E1N

D111
BAV199
–2V5A
1
A1
K2
2
+2V5A
K1/A2
3
CLAMP1P

D112
BAV199
–2V5A
1
A1
K2
2
+2V5A
K1/A2
3
CLAMP1N

C111
47p C0G 1%
AIN1P
GND

C112
47p C0G 1%
AIN1N
GND

C113
1n C0G 2%
AIN1P
AIN1N

CH2 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J102
CH2 EEG
E2P
1

R121
4.99k 0.1%
E2P
CLAMP2P
CLAMP2P
AIN2P

R122
4.99k 0.1%

R123
4.99k 0.1%
E2N
CLAMP2N
CLAMP2N
AIN2N

R124
4.99k 0.1%

R125
0R REF SELECT
REF_RAW
E2N

D121
BAV199
–2V5A
1
A1
K2
2
+2V5A
K1/A2
3
CLAMP2P

D122
BAV199
–2V5A
1
A1
K2
2
+2V5A
K1/A2
3
CLAMP2N

C121
47p C0G 1%
AIN2P
GND

C122
47p C0G 1%
AIN2N
GND

C123
1n C0G 2%
AIN2P
AIN2N

CH3 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J103
CH3 EEG
E3P
1

R131
4.99k 0.1%
E3P
CLAMP3P
CLAMP3P
AIN3P

R132
4.99k 0.1%

R133
4.99k 0.1%
E3N
CLAMP3N
CLAMP3N
AIN3N

R134
4.99k 0.1%

R135
0R REF SELECT
REF_RAW
E3N

D131
BAV199
–2V5A
1
A1
K2
2
+2V5A
K1/A2
3
CLAMP3P

D132
BAV199
–2V5A
1
A1
K2
2
+2V5A
K1/A2
3
CLAMP3N

C131
47p C0G 1%
AIN3P
GND

C132
47p C0G 1%
AIN3N
GND

C133
1n C0G 2%
AIN3P
AIN3N

CH4 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J104
CH4 EEG
E4P
1

R141
4.99k 0.1%
E4P
CLAMP4P
CLAMP4P
AIN4P

R142
4.99k 0.1%

R143
4.99k 0.1%
E4N
CLAMP4N
CLAMP4N
AIN4N

R144
4.99k 0.1%

R145
0R REF SELECT
REF_RAW
E4N

D141
BAV199
–2V5A
1
A1
K2
2
+2V5A
K1/A2
3
CLAMP4P

D142
BAV199
–2V5A
1
A1
K2
2
+2V5A
K1/A2
3
CLAMP4N

C141
47p C0G 1%
AIN4P
GND

C142
47p C0G 1%
AIN4N
GND

C143
1n C0G 2%
AIN4P
AIN4N

U1A
ADS1299PAG
AIN1P
16
IN1P
AIN1N
15
IN1N
AIN2P
14
IN2P
AIN2N
13
IN2N
AIN3P
12
IN3P
AIN3N
11
IN3N
AIN4P
10
IN4P
AIN4N
9
IN4N

Front–end RC is RF/alias prefiltering, not the 0.5–80Hz digital bandpass.
BAV199 steers to +/–2.5V; rail steering, NOT certified patient protection or a 0.3V hard clamp.
Reference leads are also current–limited through the N–side resistors.
No electrode is connected directly to board GND.

DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED
YuEEG

Sheet: /01 / Electrode inputs CH1–CH4/
File: inputs_1_4.kicad_sch

Title: 01 / Electrode inputs CH1–CH4

Size: A3	Date: 2026–10–27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 2/15

01 / ELECTRODE INPUTS CH5–CH8

Referential EEG: one SIGNAL terminal per channel. Shared REF –> all N input networks. BIAS is a separate driven electrode.

CH5 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J105
CH5 EEG
E5P
1

R151
4.99k 0.1%
R152
4.99k 0.1%
R153
4.99k 0.1%
R154
4.99k 0.1%
R155
0R REF SELECT
REF_RAW
E5N

CH6 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J106
CH6 EEG
E6P
1

R161
4.99k 0.1%
R162
4.99k 0.1%
R163
4.99k 0.1%
R164
4.99k 0.1%
R165
0R REF SELECT
REF_RAW
E6N

CH7 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J107
CH7 EEG
E7P
1

R171
4.99k 0.1%
R172
4.99k 0.1%
R173
4.99k 0.1%
R174
4.99k 0.1%
R175
0R REF SELECT
REF_RAW
E7N

CH8 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J108
CH8 EEG
E8P
1

R181
4.99k 0.1%
R182
4.99k 0.1%
R183
4.99k 0.1%
R184
4.99k 0.1%
R185
0R REF SELECT
REF_RAW
E8N

D151
BAV199
D152
BAV199
D161
BAV199
D162
BAV199
D171
BAV199
D172
BAV199
D181
BAV199
D182
BAV199
−2V5A
+2V5A
A1
K2
K1/A2
CLAMP5P
CLAMP5N
CLAMP6P
CLAMP6N
CLAMP7P
CLAMP7N
CLAMP8P
CLAMP8N

C151
47p C0G 1%
C152
47p C0G 1%
C153
1n C0G 2%
C161
47p C0G 1%
C162
47p C0G 1%
C163
1n C0G 2%
C171
47p C0G 1%
C172
47p C0G 1%
C173
1n C0G 2%
C181
47p C0G 1%
C182
47p C0G 1%
C183
1n C0G 2%
AIN5P
AIN5N
AIN6P
AIN6N
AIN7P
AIN7N
AIN8P
AIN8N
GND

U1B
ADS1299PAG
IN5P
IN5N
IN6P
IN6N
IN7P
IN7N
IN8P
IN8N

Front–end RC is RF/alias prefiltering, not the 0.5–80Hz digital bandpass.
BAV199 steers to +/–2.5V; rail steering, NOT certified patient protection or a 0.3V hard clamp.
Reference leads are also current–limited through the N–side resistors.
No electrode is connected directly to board GND.

DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED
YuEEG

Sheet: /01 / Electrode inputs CH5–CH8/
File: inputs_5_8.kicad_sch

Title: 01 / Electrode inputs CH5–CH8

Size: A3	Date: 2026–10–27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 3/15

02 / ADS1299 DIGITAL, BIAS, REFERENCE AND DECOUPLING

U1C
ADS1299PAG

ADS_MOSI 34 DIN
ADS_SCLK 40 SCLK
ADS_CS 39 CS
ADS_RESET 36 RESET
ADS_PWDN 35 PWDN
ADS_START 38 START
CLKSEL 52 CLKSEL
37 CLK
GND 41 DAISY_IN
GND 31 RESV1

DOUT 43 ADS_MISO
DRDY 47 ADS_DRDY
GPIO1 42 GPIO1_PD
GPIO2 44 GPIO2_PD
GPIO3 45 GPIO3_PD
GPIO4 46 GPIO4_PD

+3V3 R201 10k ADS_CS
GND R202 10k ADS_RESET
GND R203 10k ADS_PWDN
+3V3 R204 10k CLKSEL

ADS_START R205 10k GND
GPIO1_PD R206 10k GND
GPIO2_PD R207 10k GND
GPIO3_PD R208 10k GND
GPIO4_PD R209 10k GND

Internal 2.048MHz clock: CLKSEL pulled high; CLK output disabled.
Use SPI MODE1. DRDY -> ISR notification -> SPI/DMA task -> packet FIFO.
RESET/PWDN default LOW. Release PWDN after rails settle; hold RESET.
Wait 10s initially; verify VCAP1–AVSS >1.1V BEFORE releasing RESET.

U1E
ADS1299PAG

+2V5A 19 AVDD
+2V5A 21 AVDD
+2V5A 22 AVDD
+2V5A 56 AVDD
+2V5A 59 AVDD
+2V5A 54 AVDD1
+3V3 48 DVDD
+3V3 50 DVDD

AVSS 20 −2V5A
AVSS 23 −2V5A
AVSS 32 −2V5A
AVSS 57 −2V5A
AVSS 58 −2V5A
AVSS1 53 −2V5A
DGND 33 GND
DGND 49 GND
DGND 51 GND
VCAP1 28 VCAP1
VCAP2 30 VCAP2
VCAP3 55 VCAP3
VCAP4 26 VCAP4
NC 27
NC 29

C210..215: AVDD to AVSS at EACH analog supply pair.
C216/217: one at EACH digital supply pin.
Keep VCAP and VREF loops short.

U1D
ADS1299PAG

17 SRB1
18 SRB2
−2V5A 60 BIASREF
BIAS_INV 61 BIASINV
−2V5A 62 BIASIN

BIASOUT 63 BIAS_OUT
VREFP 24 VREFP
VREFN 25 −2V5A
RESERVED 64

R210 1M 1% BIAS_OUT BIAS_INV
R211 499k 1% BIAS_OUT BIAS_LIMIT
R212 499k 1% BIAS_LIMIT BIAS_ELECTRODE

C201 1.5n C0G BIAS_OUT BIAS_INV

J201
REF / BIAS
REF_RAW 1
BIAS_ELECTRODE 2

C210 1u 10V X7R +2V5A −2V5A
C211 1u 10V X7R +2V5A −2V5A
C212 1u 10V X7R +2V5A −2V5A
C213 1u 10V X7R +2V5A −2V5A
C214 1u 10V X7R +2V5A −2V5A
C215 1u 10V X7R +2V5A −2V5A

VREFP C202 10u 10V X7R −2V5A
VREFP C203 100n 10V −2V5A

C216 1u 10V X7R +3V3 GND
C217 1u 10V X7R +3V3 GND

BIASREF_INT=1 -> internal near–0V midpoint. SRB1/SRB2 unused.
Use CHnSET=0x60 and MISC1=0x00 (external N inputs).
Start with BIAS_SENSP=0xFF, BIAS_SENSN=0x00; validate loop stability.
BIAS feedback 1M || 1.5nF is a TI example, not a guaranteed stable loop.
Lead-off excitation disabled in baseline recording; enable only in a separate test mode.

D801
BAV199 BIAS CLAMP
−2V5A 1 A1
K2 2 +2V5A
K1/A2
3 BIAS_LIMIT

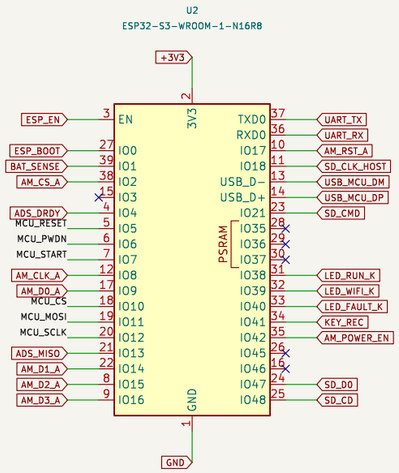
VCAP1 C220 100u 10V −2V5A
VCAP2 C221 1u 10V −2V5A
VCAP3 C222 1u 10V −2V5A

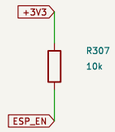
VCAP3 C223 100n 10V −2V5A
VCAP4 C224 1u 10V −2V5A

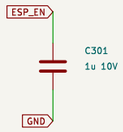
DUAL USB / AMOLED / SD BACKUP — ENGINEERING VALIDATION REQUIRED
YuEEG

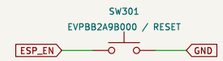
Sheet: /02 / ADS1299 digital, bias, reference and decoupling/		
File: afe_core.kicad_sch		
Title: 02 / ADS1299 digital, bias, reference and decoupling		
Size: A3	Date: 2026-10-27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 4/15

03 / ESP32-S3 WIRELESS ACQUISITION CONTROLLER











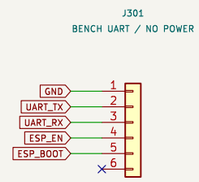




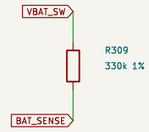


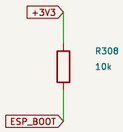






J301 is 3.3V UART, not RS-232.
No USB or charging power path is fitted.
Disconnect ALL electrodes before attaching
UART, debugger, scope or external instruments.



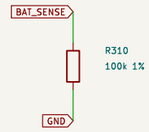


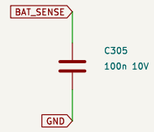












VBAT/4.3 -> GPIO1 ADC1. Max 8.4V -> 1.95V.
Calibrate ADC and stop acquisition below 6.4V.
Pack protection is external; this divider is NOT a BMS.

GPIO map: IO4 DRDY / IO5 RESET / IO6 PWDN / IO7 START / IO10 CS / IO11 MOSI / IO12 SCLK / IO13 MISO.
Boot strap pins IO3, IO45, IO46 are unused. Choose N8 (no PSRAM), not an arbitrary module variant.
Transmit int24 samples + sequence counter + sample-rate/gain metadata; target 500SPS, 24x.
Do not print volts from the interrupt. This hardware requires new pin mapping and firmware integration.
RF module antenna at PCB edge; preserve Espressif antenna keep-out on all layers.

DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED

YuEEG

Sheet: /03 / ESP32-S3 wireless acquisition controller/
File: mcu_radio.kicad_sch

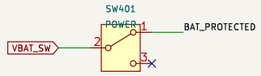
Title: 03 / ESP32-S3 wireless acquisition controller

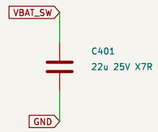
Size: A3	Date: 2026-10-27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 5/15

04 / BATTERY INPUT AND INDEPENDENT ANALOG–DIGITAL SUPPLIES

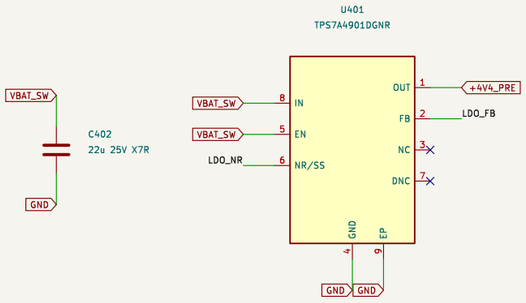


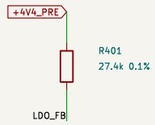


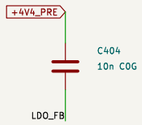


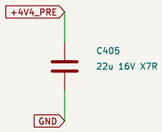


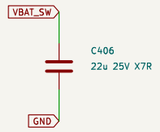
Onboard 2S charger/protection/balancer: see battery–management sheets.
J401 now has B+, MID, B–; connect a matched 2S pack plus a separate insulated NTC.
Isolated USB may supply acquisition via delayed SYS path. POWER OFF for charging. System isolation validation pending.

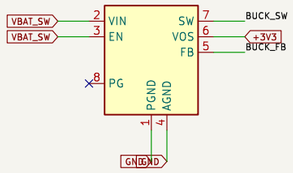




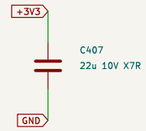


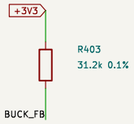


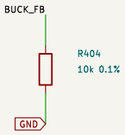


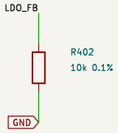


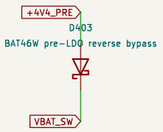




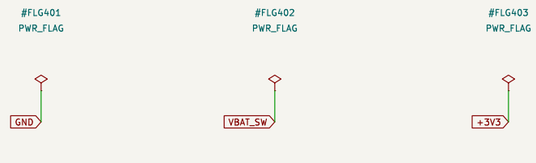








Pre–regulator (NOT ADS AVDD): 1.185 x (1 + 27.4k/10k) = 4.432V nominal.
22uF nominal input/output; >=10uF effective after DC bias.
Feeds TPS73225 and TPS60403 on bipolar supply sheet. LDO EP to GND. Keep digital switching return away from AFE.





Digital: 0.8 x (1 + 31.2k/10k) = 3.296V nominal.
VOS senses output AFTER inductor. 1A rated converter.
Minimize VIN–cap/SW/PGND hot loop; validate Wi–Fi transients.

DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED

YuEEG

Sheet: /04 / Battery input and independent analog–digital supplies/
File: power.kicad_sch

Title: 04 / Battery input and independent analog–digital supplies

Size: A3	Date: 2026–10–27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 6/15

BIPOLAR ANALOG SUPPLY: +2.5V / −2.5V; independent 3.3V digital ground reference

RESET/PWDN default LOW. After rails settle: PWDN HIGH, keep RESET LOW.
Wait >=tPOR AND VCAP1−AVSS >1.1V; use 10s initial bench delay.
100uF / internal 9.8k means seconds, NOT 200ms. Verify with scope.
Then RESET HIGH; wait >=18 CLK; SDATAC/configure/read back/start.
No automatic analog−rail/VCAP monitor is fitted; qualify timing on bench.

+4V4_PRE
C410
1u 10V X7R
GND

U403
TPS73225DBVR
+4V4_PRE
+4V4_PRE
IN
EN
GND
OUT
NR
+2V5A
POS_NR
GND

+2V5A
C412
10u 10V X7R
GND

+2V5A
R410
4.99k 1%
GND

POS_NR
C411
10n COG
GND

C414
2.2u 10V X7R
CFLY_PLUS
CFLY_MINUS

NEG_RAW
D402
BAT46W reverse−current bypass
−2V5A

+4V4_PRE
C413
2.2u 10V X7R
GND

U404
TPS60403DBVR
+4V4_PRE
IN
GND
OUT
CFLY+
CFLY−
NEG_RAW
CFLY_PLUS
CFLY_MINUS
GND

GND
C415
10u 10V X7R
NEG_RAW

U405
TPS72325DBVR
NEG_RAW
+3V3
IN
EN
GND
OUT
NR
−2V5A
NEG_NR
GND

GND
C417
10u 10V X7R
−2V5A

GND
R411
4.99k 1%
−2V5A

NEG_NR
C416
10n COG
GND

GND
C418
1u 10V X7R
NEG_RAW

TP405
+2V5A
+2V5A

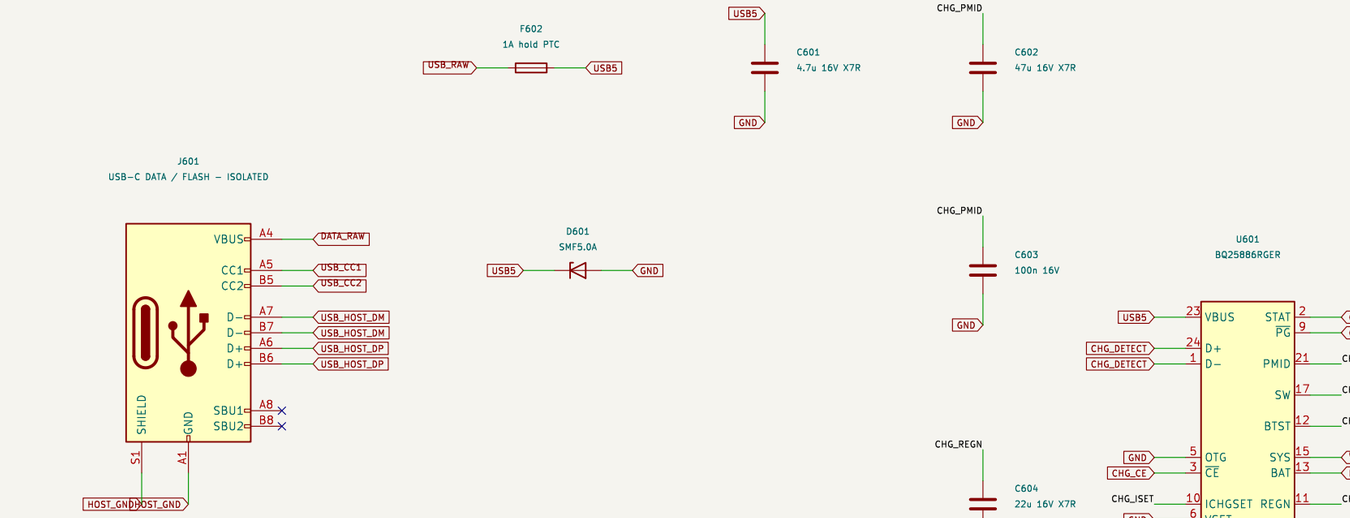
TP406
−2V5A
−2V5A

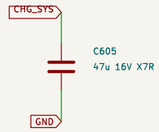
TP407
NEG_RAW
NEG_RAW

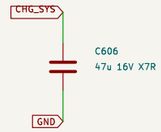
4.43V pre−regulator −> positive LDO: 250kHz inverter −> negative low−noise LDO.
Keep CFLY/input/output capacitor loops at U404. Do not route switching nodes into the AFE.
TPS723 EN is +3V3 (not +4V4_PRE); 10nF NR to GND. D402 bypasses reverse current at power−down.
4.99k rail bleeders provide ~0.5mA load/discharge; they do not absorb arbitrary electrode faults.

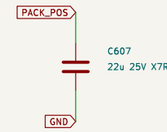
DUAL USB / AMOLED / SD BACKUP − ENGINEERING VALIDATION REQUIRED
YuEEG
Sheet: /05 / Bipolar analog supply/
File: power_bipolar.kicad_sch
Title: 05 / Bipolar analog supply and power−up requirements
Size: A3 | Date: 2026−10−27 | **Rev: E**
KiCad E.D.A. 9.0.5 | Id: 7/15

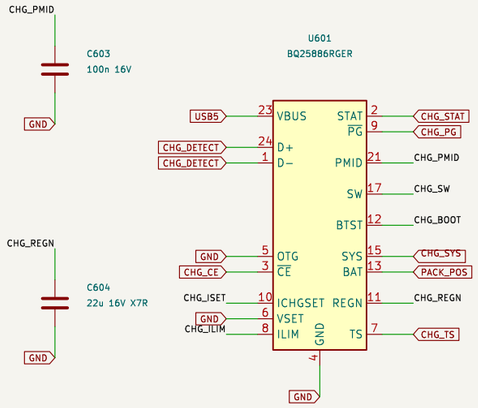
USB-C 5V INPUT / 2S Li-ion CHARGER / 300mA / 8.2V CONSERVATIVE CV



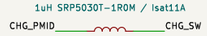




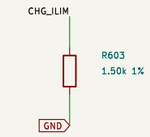


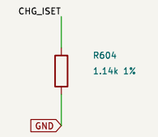


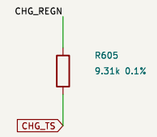


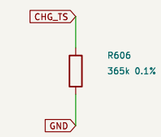




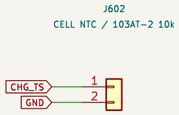


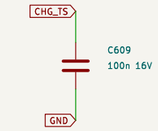


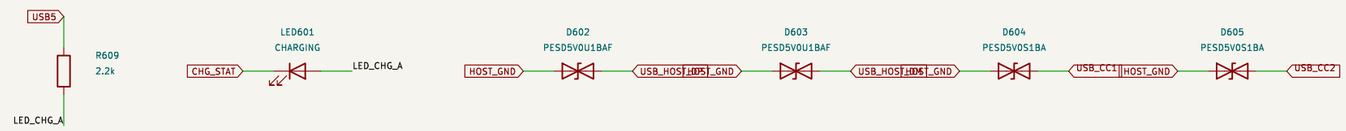


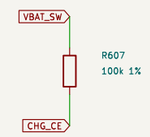


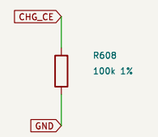














USB HOST_GND and HOST_RAW are isolated from system GND. See isolated_usb sheet.
Charger DP/DM are shorted ONLY locally; USB data uses ADuM4160 to ESP32-S3.
Charging input is enabled only for Type-C 1.5A/3A advertisement; legacy ports support battery-powered data.
8.2V / nominal 299mA; independent insulated NTC. POWER ON holds CE high: acquisition and charge are separate modes.
USB power can supply the board through charger SYS and a delayed interlocked path; no unprotected BAT bypass.

DUAL USB / AMOLED / SD BACKUP - ENGINEERING VALIDATION REQUIRED

YuEEG

Sheet: /06 / USB-C standalone 2S charger and battery NTC/
File: battery_charger.kicad_sch

Title: 06 / USB-C standalone 2S charger and battery NTC

Size: A3	Date: 2026-10-27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 8/15

RAW CELL HARNESS: B+ / MID / B− ; NEVER SHORT RAW B− TO SYSTEM GND

J401
2S RAW: B+ / MID / B−
RAW_BPLUS
RAW_MID
RAW_BMINUS

F601
1A fast fuse / 32V
RAW_BPLUS
PACK_POS

U602
S-8252AAY-M6T1U
PROT_VDD
PROT_VC
PROT_VM
VDD
VC
VM
DO
CO
VSS
DSG_GATE
CHARGE_GATE
RAW_BMINUS

U604
TPS3700DDCR RUN WINDOW
PACK_POS
RUN_UV
RUN_OV
VDD
INA+
INB−
OUTA
OUTB
GND
RUN_BASE

C640
100n 16V
PACK_POS
GND

R620
470R 1%
RAW_BPLUS
PROT_VDD

R621
470R 1%
RAW_MID
PROT_VC

C620
100n 16V
PROT_VDD
RAW_BMINUS

C621
100n 16V
PROT_VC
RAW_BMINUS

R622
2.00k 1%
GND
PROT_VM

R640
1.58M 0.1%
PACK_POS
RUN_UV

R641
100k 0.1%
RUN_UV
GND

R642
2.05M 0.1%
PACK_POS
RUN_OV

R643
100k 0.1%
RUN_OV
GND

Q601
CSD17308Q3 DSG
DSG_SOURCE
S1
S2
S3
D5-8_EP
G
BMS_DRAINS
DSG_GATE

Q602
CSD17308Q3 CHG
GND
S1
S2
S3
D5-8_EP
G
BMS_DRAINS
CHARGE_GATE

R623
0.010R 1% 0.5W
RAW_BMINUS
DSG_SOURCE

C641
10n C0G
RUN_UV
GND

R644
62k 1%
PACK_POS
RUN_BASE

R645
10k
RUN_BASE
RUN_NPN_BASE

R646
1M
RUN_NPN_BASE
GND

D606
BZT52C8V2S
DSG_GATE
DSG_SOURCE

D607
BZT52C8V2S
CHARGE_GATE
GND

Q605
MMBT3904 ENABLE
RUN_PULLDOWN
RUN_NPN_BASE
GND

Q606
MMBT3904 USB INHIBIT
RUN_BASE
USB_INHIBIT_B
GND

R647
47k
USB_RAW
USB_INHIBIT_B

R648
1M
USB_INHIBIT_B
GND

Q603
AO3401A BACK-TO-BACK
PACK_POS
RUN_GATE
RUN_SOURCES

Q604
AO3401A BACK-TO-BACK
ACQ_INPUT
RUN_GATE
RUN_SOURCES

R649
1M
RUN_SOURCES
RUN_GATE

R650
100k
RUN_GATE
RUN_PULLDOWN

C642
10n 25V
RUN_GATE
RUN_SOURCES

R651
47k
ACQ_INPUT
GND

#FLG610
PWR_FLAG
RAW_BPLUS

#FLG611
PWR_FLAG
RAW_BMINUS

#FLG612
PWR_FLAG
PACK_POS

#FLG613
PWR_FLAG
PROT_VDD

Primary protection per cell: OV 4.25V / UV 3.00V; inhibit OV−cell recharge. Charge/discharge overcurrent use VM and actual series resistance.
Q601/Q602 are common−DRAIN NMOS. Q603/Q604 are common−SOURCE PMOS: no body−diode bypass when OFF.
RUN typical falling cutoff ~6.63V, restart ~6.72V; OV cutoff ~8.6V. USB clamps RUN_BASE independently of MCU.
Raw B− returns only to protection/balance circuits. USB, charger, NTC and EEG use protected GND.
Hardware power−off is NOT galvanic or patient isolation. Remove electrodes before inserting USB.

DUAL USB / AMOLED / SD BACKUP − ENGINEERING VALIDATION REQUIRED
YuEEG
Sheet: /07 / Raw 2S cell protection and fail−safe acquisition switch/
File: battery_protection.kicad_sch

Title: 07 / 2S protection and acquisition inhibit

Size: A3	Date: 2026−10−27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 9/15

CELL BALANCING: ONLY WITH VALID USB SOURCE AND RAW PACK ABOVE ~7.6V

RAW_BPLUS
R630
160R 1%
BAL_VC2

RAW_BPLUS
R633
100R 1%
BAL_VDD

BAL_VDD
C630
100n 16V
RAW_BMINUS

U603
BQ29209DRBR
BAL_VDD 7 VDD
BAL_VC2 1 VC2
BAL_VC1 2 VC1
BAL_VC1_CB 3 VC1_CB
BAL_ENABLE 6 CB_EN
OUT 8 BAL_OV_ALARM
CD 4 BAL_CD
GND 5
EP 9
RAW_BMINUS

RAW_MID
R631
260R 1%
BAL_VC1

BAL_VC2
C631
100n 16V
BAL_VC1

U605
TPS3700DDCR BAL UV
RAW_BPLUS 5 VDD
BAL_UV_SENSE 3 INA+
RAW_BMINUS 4 INB−
OUTA 1 RAW_OK
OUTB 6
GND 2
RAW_BMINUS

RAW_MID
R632
100R 1%
BAL_VC1_CB

BAL_VC1
C632
100n 16V
RAW_BMINUS

RAW_BPLUS
R660
1.81M 0.1%
BAL_UV_SENSE

BAL_UV_SENSE
R661
100k 0.1%
RAW_BMINUS

RAW_BPLUS
R662
1M
RAW_OK

RAW_BPLUS
C660
100n 16V
RAW_BMINUS

BAL_CD
C633
330n 16V
RAW_BMINUS

BAL_VDD
R634
330k
BAL_ENABLE

USB5
R663
2.2k
BAL_OPTO_A

U606
LTV−356T VALID USB
BAL_OPTO_A 1
CHG_PG 2
4 BAL_ENABLE
3 BAL_AND_SOURCE

Q607
2N7002 RAW UV GATE
BAL_AND_SOURCE
RAW_OK 1
3
2
RAW_BMINUS

TP601
RAW_BPLUS
RAW_BPLUS

TP602
RAW_MID
RAW_MID

TP603
RAW_BMINUS
RAW_BMINUS

TP604
BAL_OV_ALARM
BAL_OV_ALARM

#FLG620
PWR_FLAG
BAL_VDD

TP605
CHG_PG
CHG_PG

TP606
CHG_STAT
CHG_STAT

BQ29209 internal balancing: TI Fig9 resistor network, about 10mA; matched cells required. Not a cell equalizer for mismatched/old cells.
VC2 capacitor returns to VC1; VC1 capacitor returns to RAW B−. Do not return raw-domain nodes to system GND.
CB_EN active LOW: 330k pull-up disables by default. Optocoupler needs /PG LOW AND Q607 enabled by raw pack UV comparator.
Raw UV typical rising ~7.64V/falling ~7.53V; prevents balancing depleted cells. Optocoupler is domain translation, NOT patient isolation.
OUT is an alarm test point only: primary charge/discharge cutoff is independently provided by S−8252AAY.
J401 sequence and external NTC wiring must be verified before cells are plugged in. No standalone charger firmware is needed.

DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED
YuEEG
Sheet: /08 / Cell balancing with raw−domain UV qualification/
File: battery_balance.kicad_sch
Title: 08 / Cell balancing with raw−domain UV qualification
Size: A3 | Date: 2026−10−27 | **Rev: E**
KiCad E.D.A. 9.0.5 | Id: 10/15

HOST SIDE: NO ELECTRODE / SYSTEM GROUND CONNECTION

F701
1.5A hold PTC
HOST_RAW
HOST_5V

DATA_5V
C701
100n 16V
HOST_GND

USB_HOST_DP
R701
24R 1%
USB_UP_DP

U701
ADuM4160BRIZ / enhanced creepage
DATA_5V
+3V3
USB_VDD1
USB_UP_DP
USB_UP_DM
USB_VDD1
USB_VDD1
HOST_GND
GND
USB_DOWN_DP
USB_DOWN_DM

USB_DOWN_DP
R703
24R 1%
USB_MCU_DP

+3V3
C703
100n 16V
GND

USB_VDD1
C702
100n 16V
HOST_GND

#FLG703
PWR_FLAG
USB_VDD1

USB_HOST_DM
R702
24R 1%
USB_UP_DM

USB_DOWN_DM
R704
24R 1%
USB_MCU_DM

D701
SMF5.0A
HOST_5V
HOST_GND

U702
TLV75533PDBVR HOST ONLY
HOST_5V
HOST_5V
IN
EN
OUT
GND
HOST_3V3
HOST_GND

HOST_5V
C704
1u 16V
HOST_GND

HOST_3V3
C705
1u 10V
HOST_GND

U703
TUSB320LAIRWBR / UFP GPIO
HOST_3V3
CHARGE_CC1
CHARGE_CC2
CC_VBUS_DET
HOST_GND
HOST_GND
VDD
CC1
CC2
VBUS_DET
PORT_UFP
EN_N
GND
OUT1
OUT2
OUT3
ID
ADDR_GPIO
CC_HIGH_N
CC_MODE2
HOST_GND

U704
SN74LVC1G04DBVR
HOST_3V3
CC_HIGH_N
VCC
NC
A
GND
Y
HOST_POWER_EN
HOST_GND

U705
TPS2553DBVR
HOST_5V
HOST_POWER_EN
HOST_ILIM
IN
EN
ILIM
GND
OUT
FAULT
HOST_DCDC_IN
HOST_GND

HOST_DCDC_IN
C708
4.7u 16V
HOST_GND

HOST_5V
C709
100n 16V
HOST_GND

#FLG701
PWR_FLAG
HOST_5V

#FLG702
PWR_FLAG
HOST_GND

HOST_5V
R705
887k 1%
CC_VBUS_DET

HOST_3V3
C706
100n 16V
HOST_GND

HOST_3V3
C707
100n 16V
HOST_GND

HOST_POWER_EN
R707
100k
HOST_GND

HOST_ILIM
R706
23.7k 1%
HOST_GND

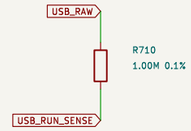
U706
THM 6-0511 / standard pinout
HOST_DCDC_IN
HOST_GND
HOST_GND
+VIN
-VIN
-VIN
+VOUT
-VOUT
USB_RAW
GND

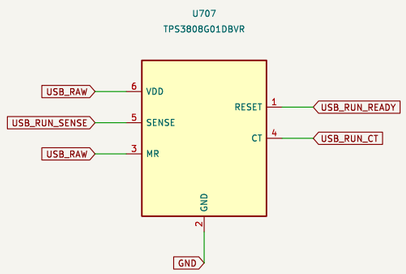
USB_RAW
C710
10u 25V X7R
GND

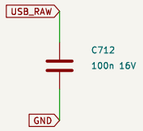
ADuM4160 BRIZ: full-speed 12Mbps; both speed pins HIGH. 24R series on all four USB lines.
Primary CC controller replaces external Rd resistors. OUT1 LOW only for advertised 1.5A/3A.
TPS2553 RILIM=23.7k: max about 1.17A before resistor tolerance; no PD negotiation.
THM6 standard pinout only, not -B1/W1 substitutions. Data isolator and converter ratings do not certify the assembled device.
No copper, vias, shield bond, external cable or screen supply may bridge HOST_GND and system GND.

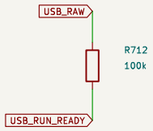
DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED
YuEEG
Sheet: /09 / Isolated USB data and conditional isolated power/
File: isolated_usb.kicad_sch
Title: 09 / Isolated USB data and conditional isolated power
Size: A3 | Date: 2026-10-27 | **Rev: E**
KiCad E.D.A. 9.0.5 | Id: 11/15

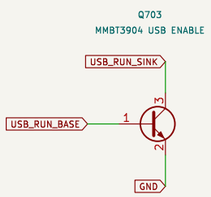
USB DATA + WIFI MAY RUN TOGETHER; CHARGING IS DISABLED WHILE POWER IS ON

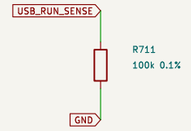




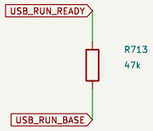


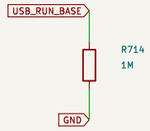


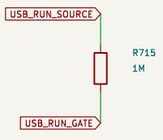


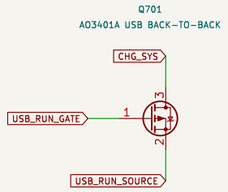


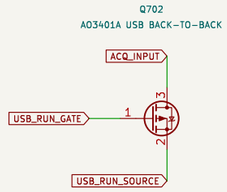


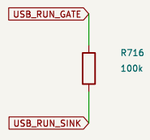


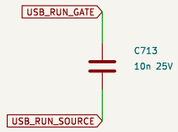












USB_RAW rising above about 4.455V must remain valid for about 0.2s before USB SYS path turns ON.
Existing battery path is inhibited immediately by Q606 sensing isolated USB_RAW. Delayed USB turn-on avoids a parallel charge path.
USB_RAW falling trips supervisor while Q606 still inhibits the battery; confirm break-before-make on bench.
POWER OFF charges battery; POWER ON supplies MCU/AFE from SYS and holds charger CE high.
Legacy/default-current USB: converter OFF, data works from battery with POWER ON. Type-C 1.5A/3A permits USB-powered flashing without battery.
USB-mode minimum pre-LDO headroom and switching noise need validation; external grounded instruments can defeat isolation.

DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED

YuEEG

Sheet: /10 / USB SYS power path with delayed battery interlock/
File: usb_system_power.kicad_sch

Title: 10 / USB SYS power path with delayed battery interlock

Size: A3	Date: 2026-10-27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 12/15

12 / AMOLED 1.91 INCH RM67162 – DWO DO0200PFS12

U1001
TPS22918DBVR AMOLED POWER
+3V3
AM_POWER_EN
AM_CT
VIN
ON
CT
VOUT
QOD
GND
AM_RAW
AM_QOD
C1003
100n 10V
C1002
22u 10V X7R
U1002
TPS7A2028PDBVR
AM_RAW
AM_2V8
IN
EN
OUT
NC
C1004
1u 10V
C1005
4.7u 10V
C1008
100n 10V
U1004
SN74AXC8T245PWR / 3V3 -> 1V8
AM_1V8
C1009
100n 10V
U1003
TPS7A2018PDBVR
C1006
1u 10V
C1007
4.7u 10V
R1001
100k
C1001
1n 25V
R1003
1k
C1010
22u 10V X7R
C1011
1u 10V
C1012
100n 10V
Q1001
MMBT3904 OE ENABLE
AM_OE_N
AM_OE_BASE
R1004
100k
R1002
10k
R1006
10k
R1005
1M
J1001
DO0200PFS12 18-PIN HARNESS
AM_IM1
R1031
47k
R1032
47k
R1033
47k
R1034
47k
R1035
47k
R1036
47k
R1037
47k
R1021
22R
R1022
22R
R1023
22R
R1024
22R
R1025
22R
R1026
22R
R1027
22R

Lock RM67162 version; module R4=0R, R3=NC, IM1 high for QSPI. VBAT=AM_RAW; VCC=2.8V; IOVCC=1.8V.
Touch held in RESET; touch signals and TE unused. 18-core harness pin numbering must match vendor CN1.
Keep screen and harness in floating system domain. Driver starts at 10MHz; refresh limited to status regions.
Dedicated SPI3 QSPI; ADC uses SPI2; microSD uses separate 1-bit SDMMC. Supplies, translation and inrush require bench tests.

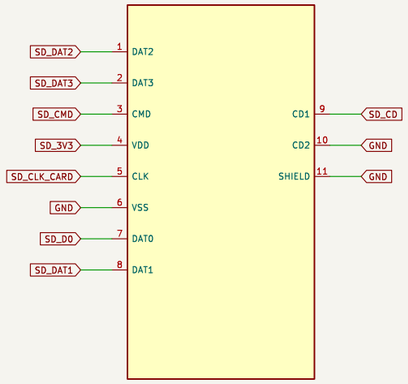
DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED
YuEEG
Sheet: /12 / AMOLED 1.91 inch RM67162 – DWO DO0200PFS12/
File: amoled_ui.kicad_sch
Title: 12 / AMOLED 1.91 inch RM67162 – DWO DO0200PFS12
Size: A3 | Date: 2026-10-27 | **Rev: E**
KiCad E.D.A. 9.0.5 | Id: 13/15

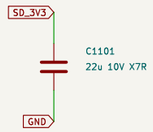
13 / SDMMC BACKUP, STATUS AND LOW-PROFILE CONTROLS

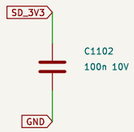


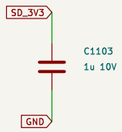
FB1101
BLM18PG121SN1D / 120R@100MHz

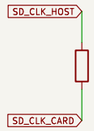
J1101
HIROSE DM3AT-SF-PEJM5



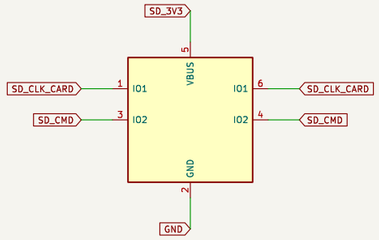




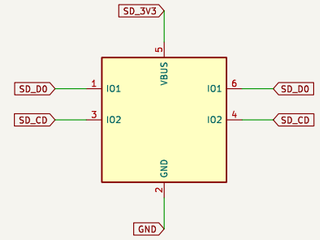


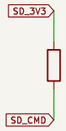


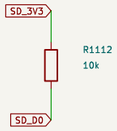
D1101
USBLC6-2SC6

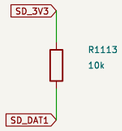


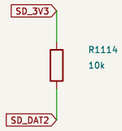
D1102
USBLC6-2SC6

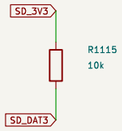


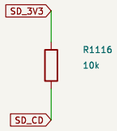






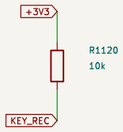


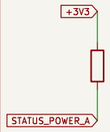


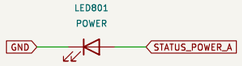


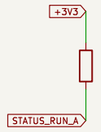
#FLG1101
PWR_FLAG



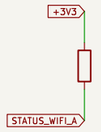




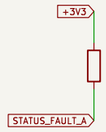




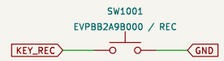














H1001
AMOLED M2 / 6mm STANDOFF



H1002
AMOLED M2 / 6mm STANDOFF



H1003
AMOLED M2 / 6mm STANDOFF



H1004
AMOLED M2 / 6mm STANDOFF



1-bit SDMMC: CMD and DAT0..3 each have 10k pull-ups, including unused DAT1/2/3. CLK has source series termination.
No SD, display or network IO in DRDY ISR. Separate recording ring and explicit drop counters; no lossless claim.
REC key toggles EEG start/stop after debounce. BOOT/RESET remain hardware buttons. All are 0.53mm low-profile metal dome switches.
AMOLED model is an assembly envelope above 6mm insulated standoffs; check cable contacts and mechanical stack with actual parts.

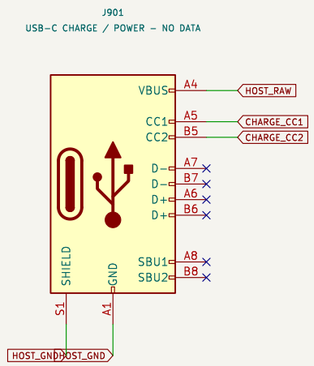
DUAL USB / AMOLED / SD BACKUP — ENGINEERING VALIDATION REQUIRED
YuEEG

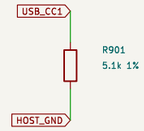
Sheet: /13 / SDMMC backup, status and low-profile controls/
File: storage_status.kicad_sch

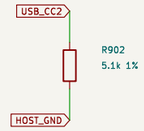
Title: 13 / SDMMC backup, status and low-profile controls

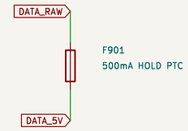
Size: A3	Date: 2026-10-27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 14/15

14 / SEPARATE USB-C DATA AND CHARGE PRIMARY-SIDE PORTS

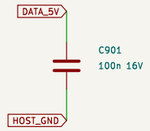


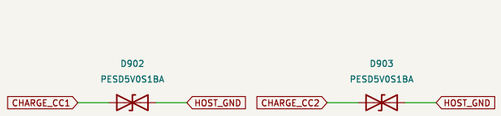














J601 DATA supplies only ADuM upstream side through F901. It cannot charge or power the acquisition domain.
J901 CHARGE supplies existing F701 / CC current detector / isolated DC-DC chain. D+/D- are NC.
No DATA_RAW/DATA_5V to HOST_RAW/HOST_5V connection or diode OR. Both ports share only HOST_GND and shields.
DATA-only requires battery; CHARGE 1.5A/3A + POWER ON can supply acquisition with charging disabled. POWER OFF charges.
Do not replace the isolation chain by a non-isolated ESP dev-board power circuit.

DUAL USB / AMOLED / SD BACKUP - ENGINEERING VALIDATION REQUIRED
YuEEG

Sheet: /14 / Separate USB-C DATA and CHARGE primary-side ports/		
File: dual_usb_ports.kicad_sch		
Title: 14 / Separate USB-C DATA and CHARGE primary-side ports		
Size: A3	Date: 2026-10-27	**Rev: E**
KiCad E.D.A. 9.0.5		Id: 15/15